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Computer Vision

Chapter 7 (part 1): Object recognition

Contents

- Overview of 'semantic vision'?
- Image classification/ recognition
- Bag-of-words
 - Recall
 - Vocabulary tree
- Classification
 - K nearest neighbors
 - Naïve Bayes
 - [Support vector machine]



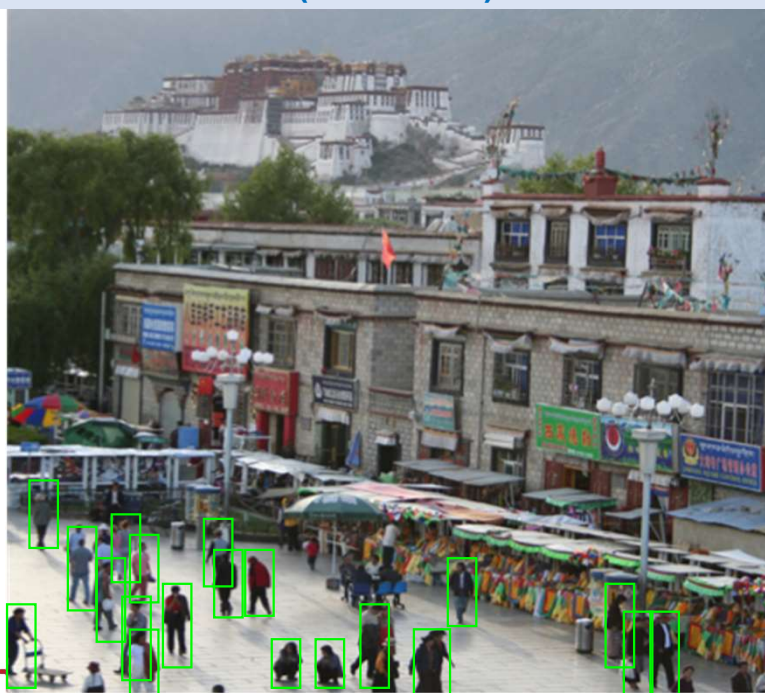
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Is this a street light?
(Recognition / classification)



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Where are the people?
(Detection)



4

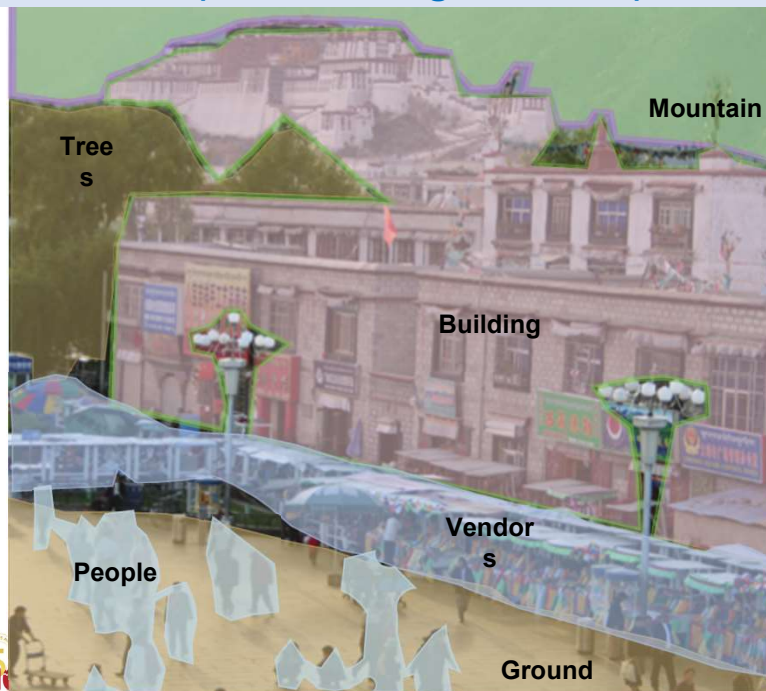
Is that Potala palace? (Identification)



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What's in the scene? (semantic segmentation)



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What type of scene is it?
(Scene categorization)



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How many object categories are there?



Challenge: variable viewpoint



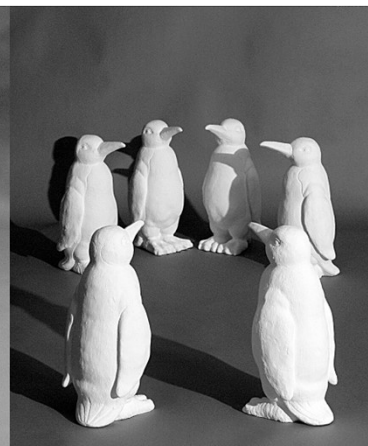
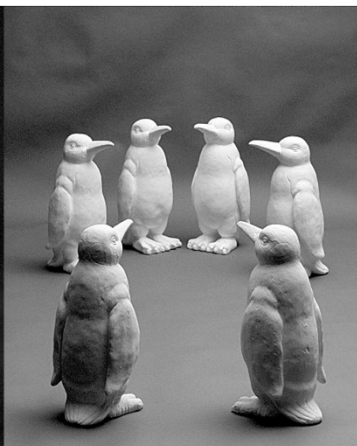
Michelangelo 1475-1564



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Challenge: variable illumination



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image credit: J. Koenderink

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Challenge: scale

and small things
from Apple.
(Actual size)



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Challenge: deformation



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Challenge: Occlusion



Magritte, 1957



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Challenge: background clutter



Kilmeny Niland. 1995



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Challenge: intra-class variations



Image Classification/ Recognition



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Image Classification/ Recognition



(assume given set of discrete labels)
{dog, cat, truck, plane, ...}



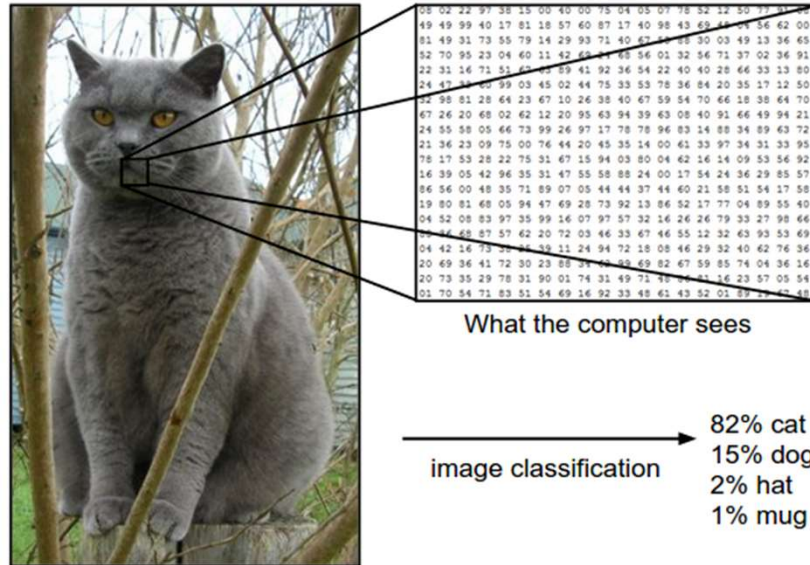
cat



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Image Classification: Problem



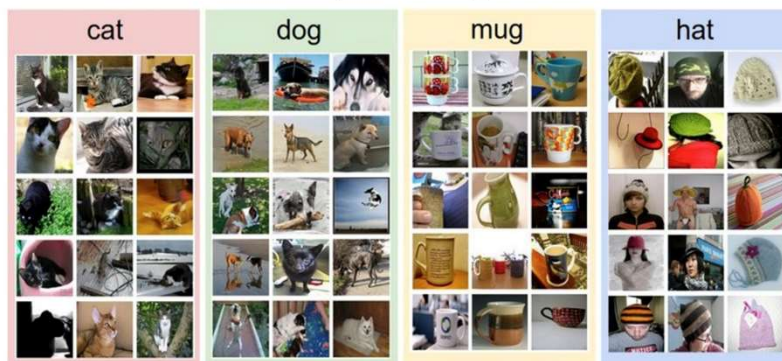
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Data-driven approach

- Collect a database of images with labels
- Use ML to train an image classifier
- Evaluate the classifier on test images

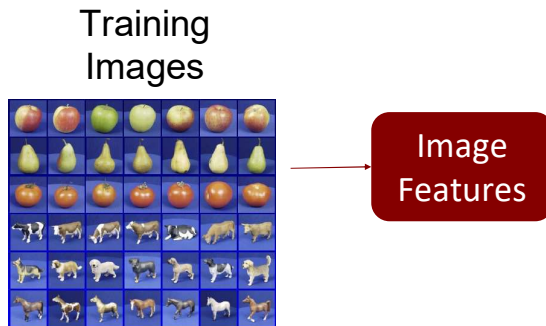
Example training set



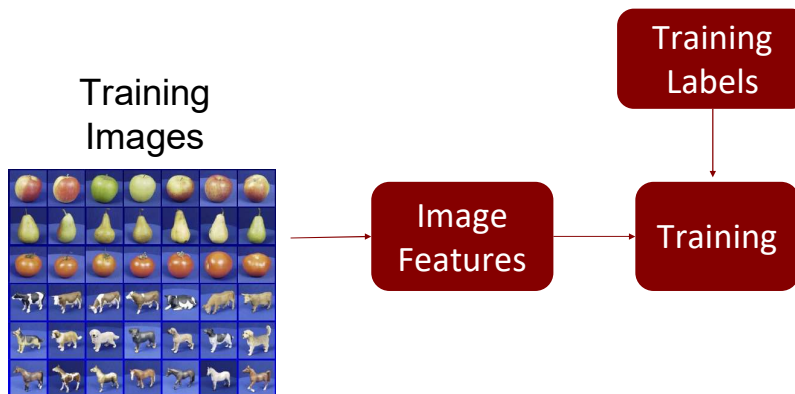
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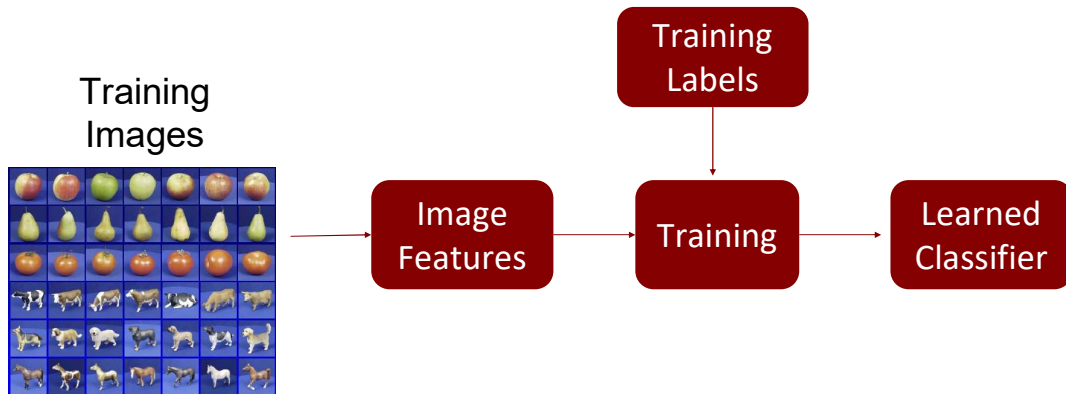
A simple pipeline - Training



A simple pipeline - Training



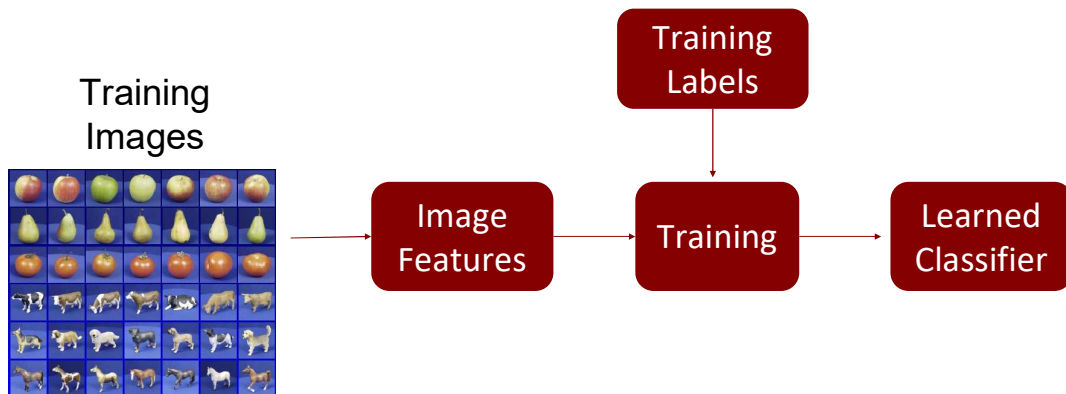
A simple pipeline - Training



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A simple pipeline - Training



Test Image



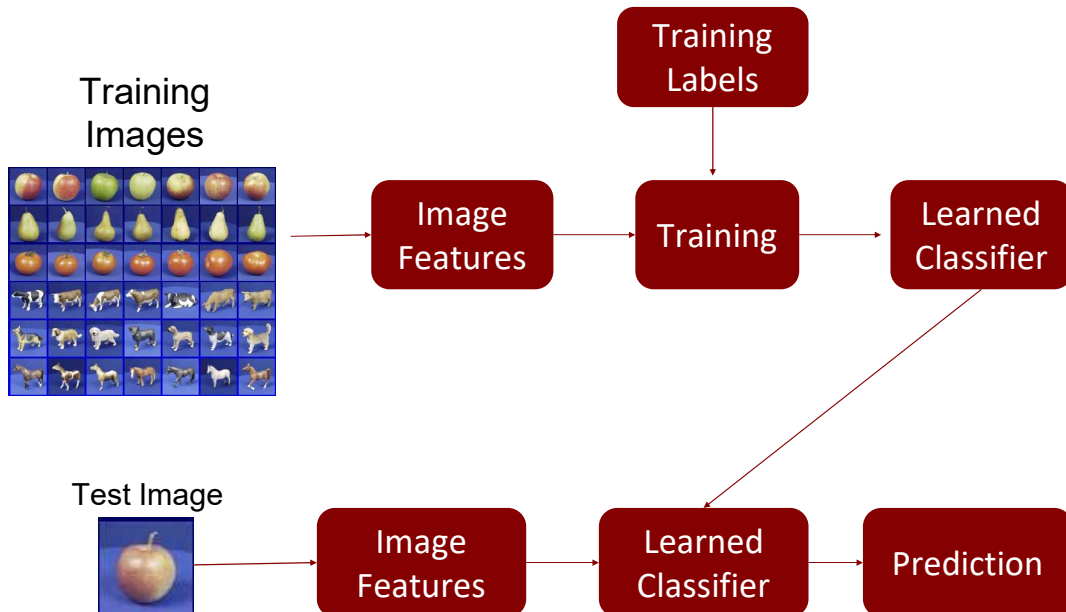
Image Features



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A simple pipeline - Training



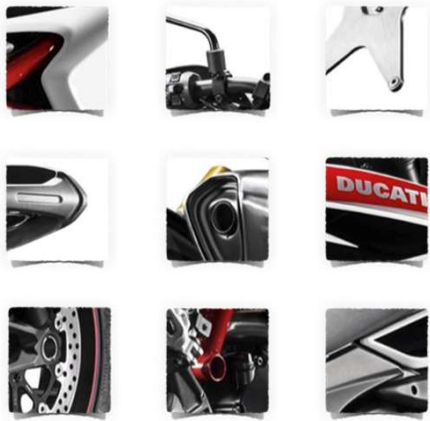
Bag of words

Basic model

Vocabulary tree

Some local feature are very informative

An object as

a collection of local features
(bag-of-features)

- deals well with occlusion
- scale invariant
- rotation invariant

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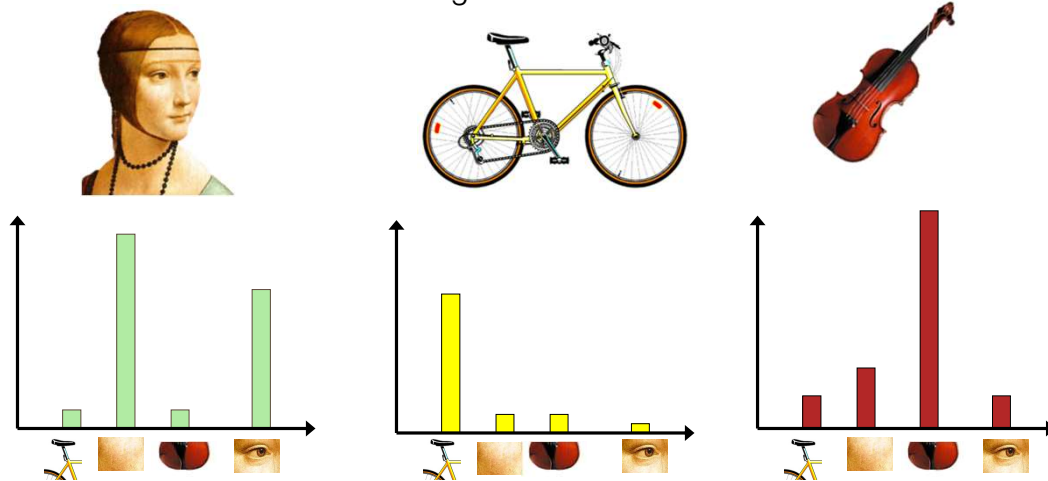
Bag-of-words

- Local feature ~~ a word
- An image ~~ a document
- Apply a technique for textual document representation: vector model



Bag-of-words

represent a data item (document, texture, image)
as a histogram over features



Standard BOW pipeline

(for image classification)

Dictionary Learning:

Learn Visual Words using clustering

Encode:

build Bags-of-Words (BOW) vectors
for each image

Classify:

Train and test data using BOWs



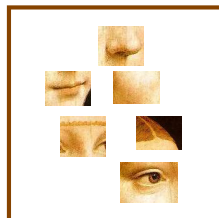
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Dictionary Learning:

Learn Visual Words using clustering

1. **extract features** (e.g., SIFT) from images

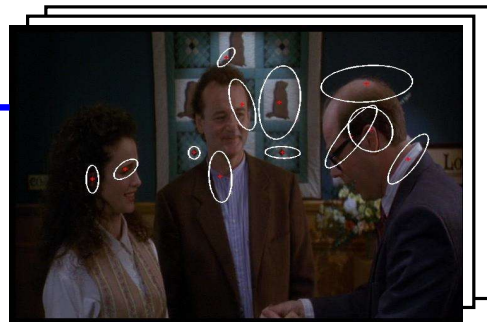
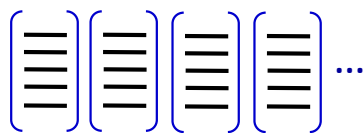
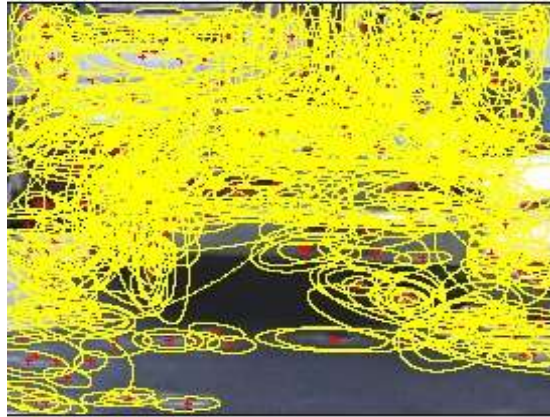


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What kinds of features can we extract?

- Regular grid
 - Vogel & Schiele, 2003
 - Fei-Fei & Perona, 2005
- Interest point detector
 - Csurka et al. 2004
 - Fei-Fei & Perona, 2005
 - Sivic et al. 2005
- Other methods
 - Random sampling (Vidal-Naquet & Ullman, 2002)
 - Segmentation-based patches (Barnard et al. 2003)

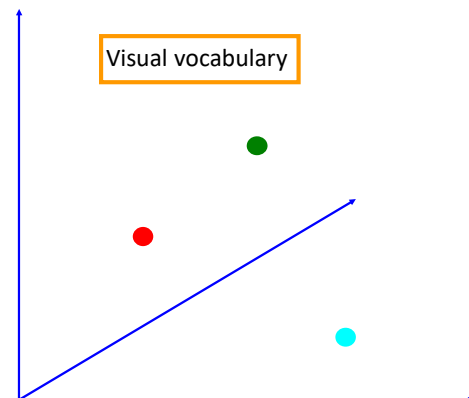
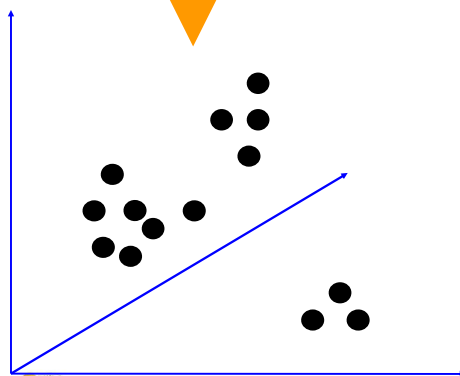
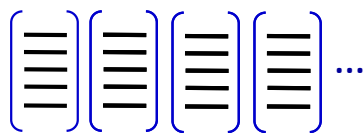


Dictionary Learning: Learn Visual Words using clustering

2. Learn visual dictionary (e.g., K-means clustering)



How do we learn the dictionary?



Clustering

K-means clustering

Example dictionary



Appearance codebook



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Source: B. Leibe

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Dictionary Learning:

Learn Visual Words using clustering

Encode:

build Bags-of-Words (BOW) vectors
for each image

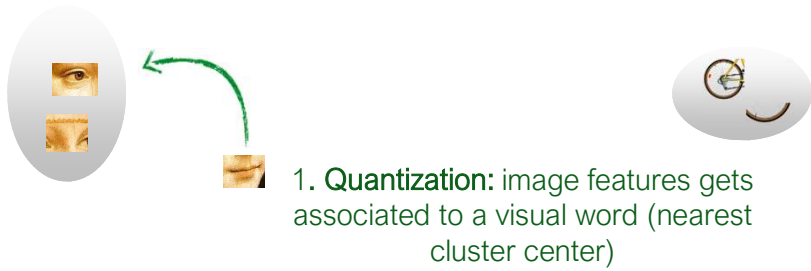
Classify:

Train and test data using BOWs



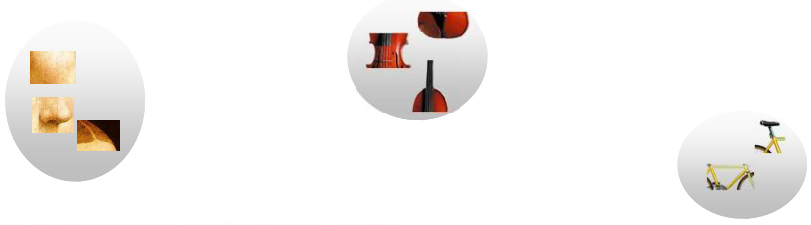
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1. **Quantization:** image features gets associated to a visual word (nearest cluster center)

Encode:
build Bags-of-Words (BOW) vectors for each image

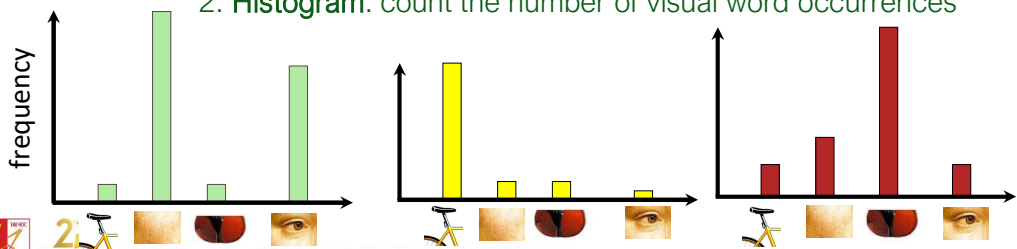


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Encode:
build Bags-of-Words (BOW) vectors for each image

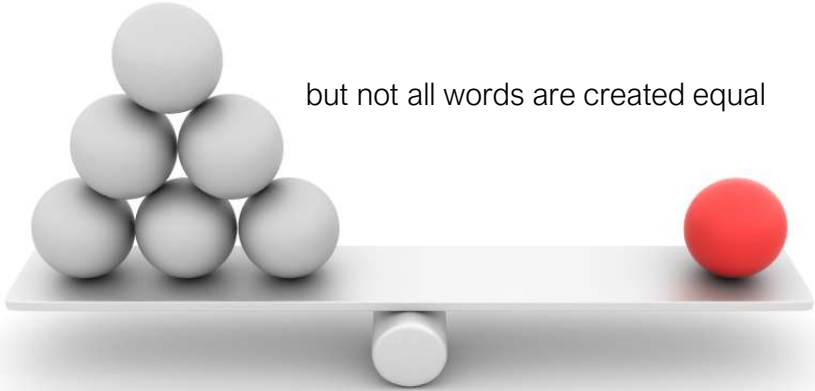
2. **Histogram:** count the number of visual word occurrences



frequency

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but not all words are created equal

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TF-IDF

Term **F**requency Inverse Document **F**requency

$$\mathbf{v}_d = [n(w_{1,d}) \quad n(w_{2,d}) \quad \cdots \quad n(w_{T,d})]$$

weight each word by a heuristic

$$\mathbf{v}_d = [n(w_{1,d})\alpha_1 \quad n(w_{2,d})\alpha_2 \quad \cdots \quad n(w_{T,d})\alpha_T]$$

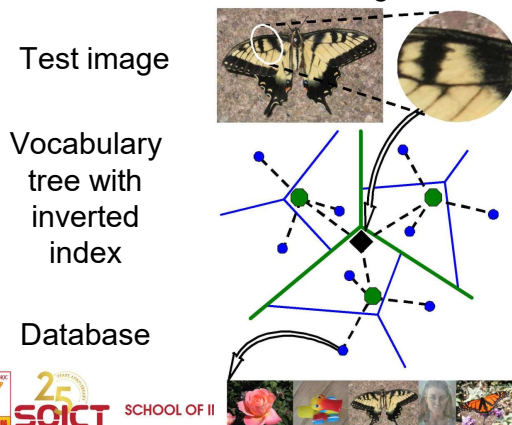
$$n(w_{i,d})\alpha_i = \overset{\substack{\text{term} \\ \text{frequency}}}{n(w_{i,d})} \log \left\{ \overset{\substack{\text{inverse document} \\ \text{frequency}}}{\frac{D}{\sum_{d'} \mathbf{1}[w_i \in d']}} \right\}$$

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Scalability: Alignment to large databases

- What if we need to align a test image with thousands or millions of images in a model database?
 - Efficient putative match generation
 - Fast nearest neighbor search, inverted indexes



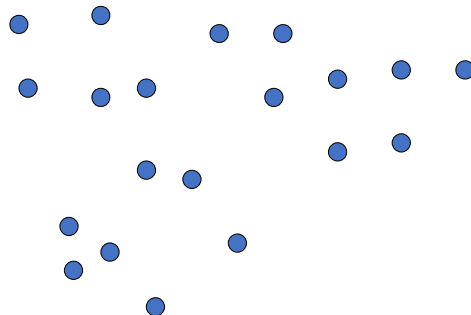
D. Nistér and H. Stewénus, [Scalable Recognition with a Vocabulary Tree](#), CVPR 2006

slide: S. Lazebnik

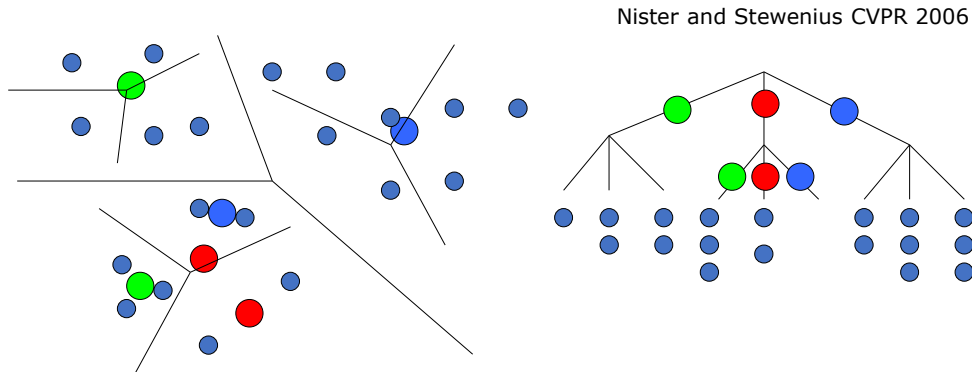
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What is a Vocabulary Tree?

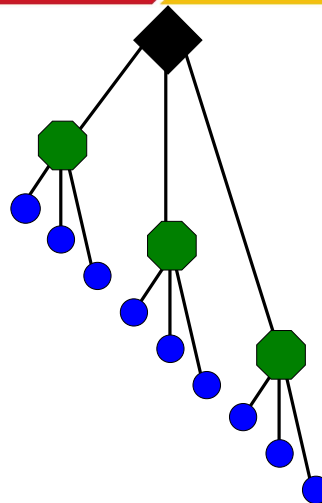
Nister and Stewenius CVPR 2006



What is a Vocabulary Tree?

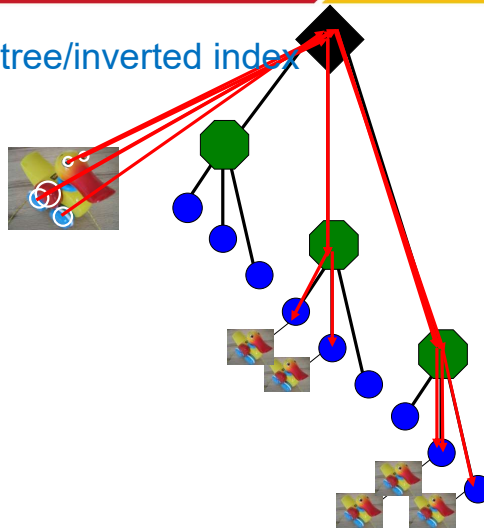


- Multiple rounds of K-Means to compute decision tree (offline)
- Fill and query tree online



Populating the vocabulary tree/inverted index

Model images

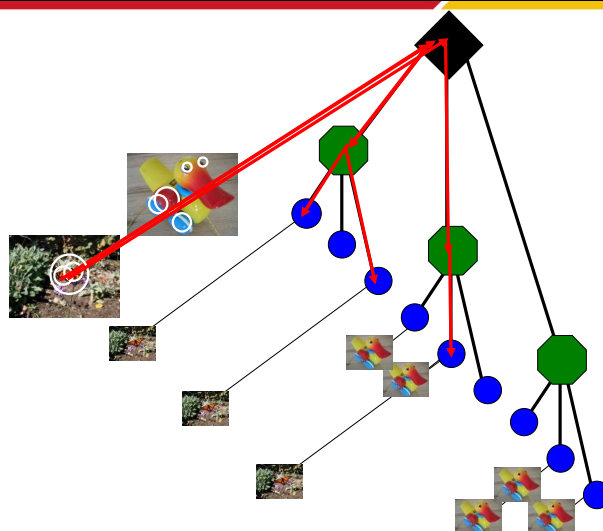


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Slide credit: D. Nister

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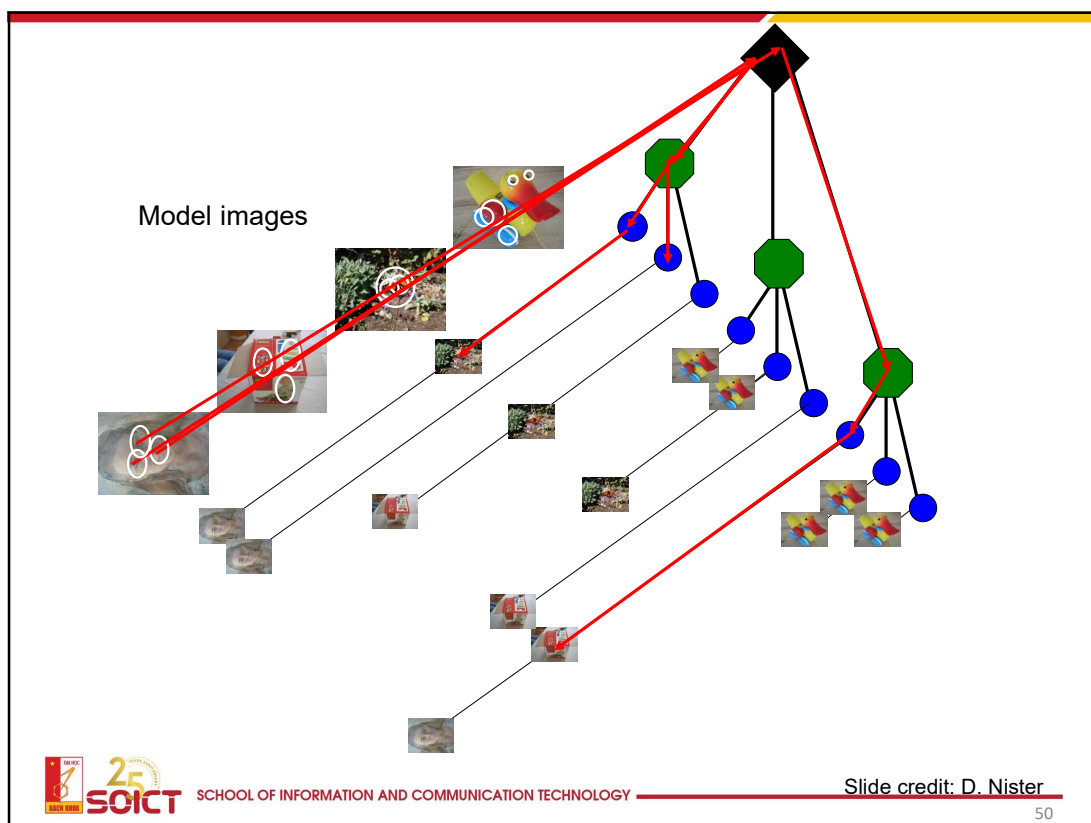
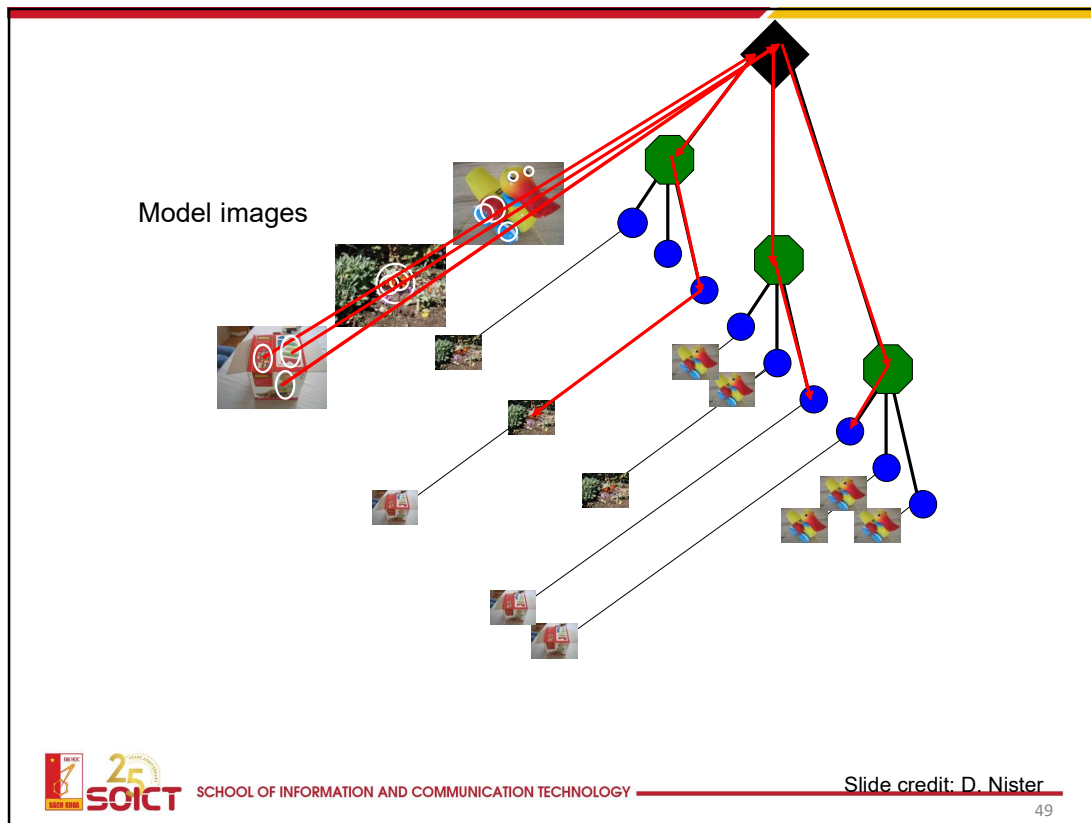
Model images

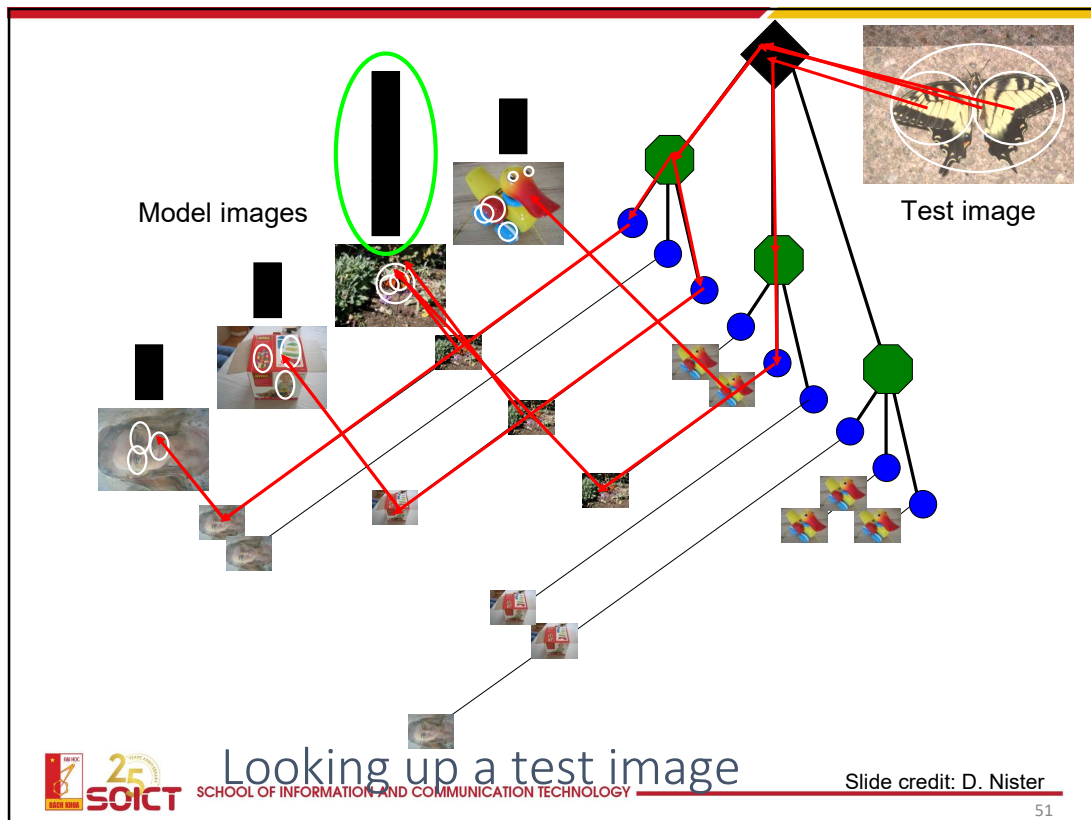


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Slide credit: D. Nister

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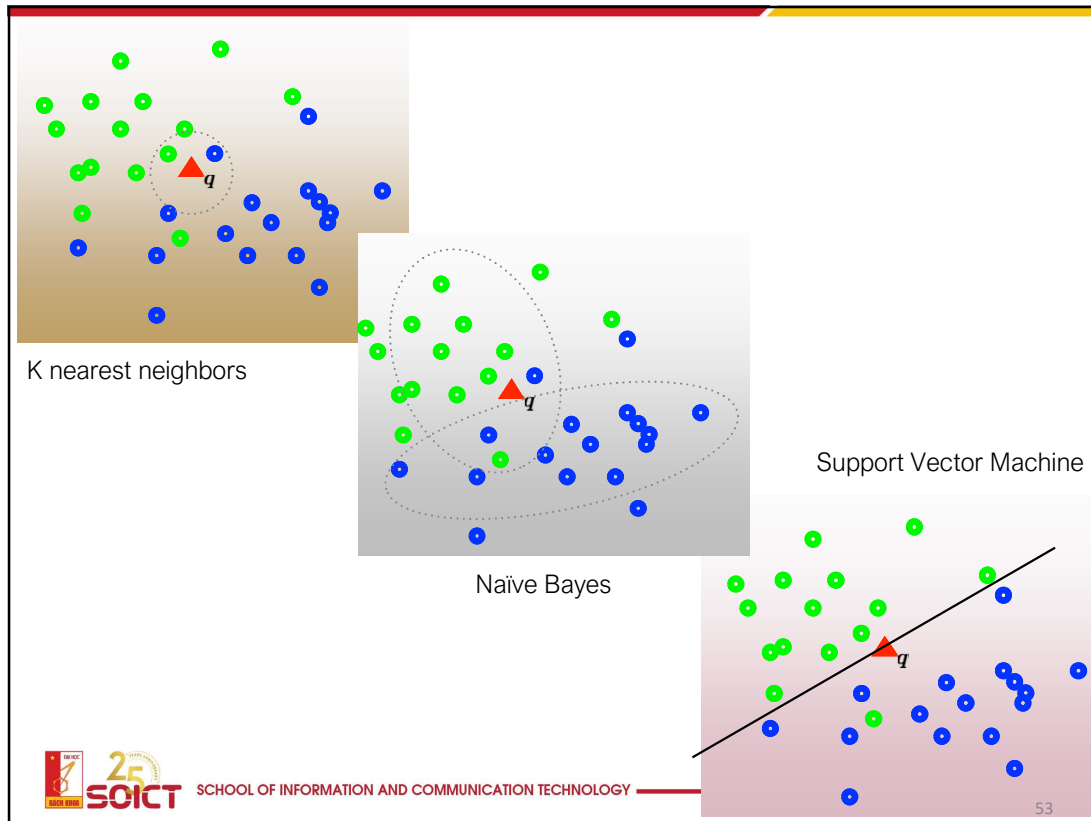




Dictionary Learning:
Learn Visual Words using clustering

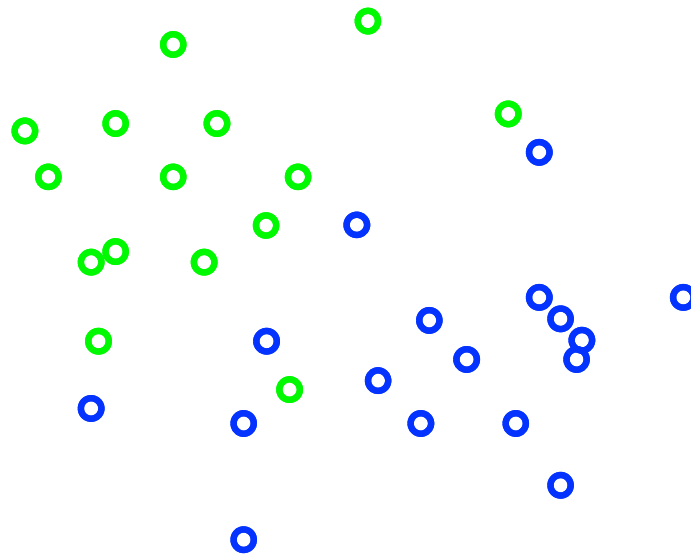
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Classify:
Train and test data using BOWs

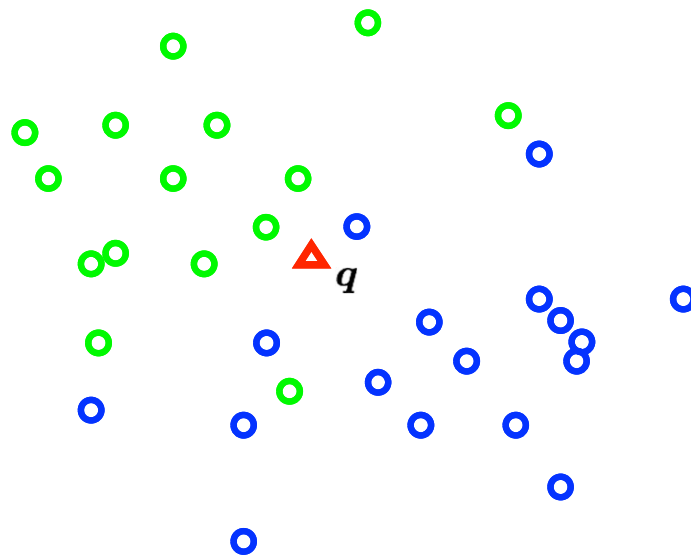


K nearest neighbors

Distribution of data from two classes

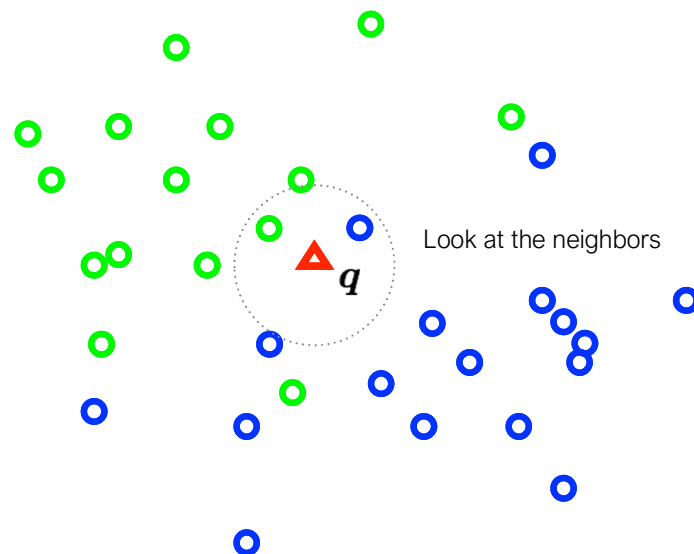


Distribution of data from two classes

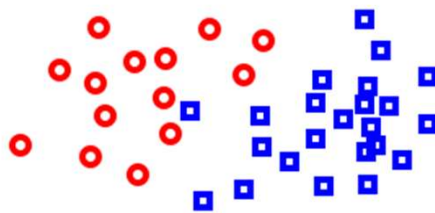


Which class does q belong too?

Distribution of data from two classes



K-Nearest Neighbor (KNN) Classifier

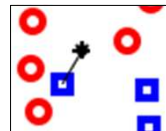


Non-parametric pattern classification approach

Consider a two class problem where each sample consists of two measurements (x,y) .

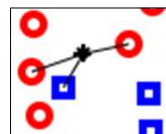
For a given query point q , assign the class of **the nearest neighbor**

$k = 1$



Compute the **k nearest neighbors** and assign the class by majority vote.

$k = 3$



Nearest Neighbor is competitive

40281508803272364955729784650087671127400726386420140578214711366
 50711106679664143112241082634004230017113109975414895351982239401029
 468681467933943144705160444612336459685605641865284554770782237018
 76953465018228357808571101378507110114527623020546972136418240510226
 9347771490618421281007833137613160574754958491501320748220251514889
 8204946233564809283671460491280070911677914592504108908989425798980
 35517816913995516228671460403321368985835452056383995196671313660901
 94368160413174151001162198405649071651525185470670258104571851200607
 6857389386823977629286168879180172075140209862393802111142972512199
 1485343471507403153959369036398241286855394426514414435912233029009
 931909754120105149336152522026601203025579580895032570884588454549
 6928545794921834078393405623926006128752047750564674307507426819404
 1284527811303570319363173084826529459091642972116797590821445761325
 906623677286083029832539300195139601414122794939282718091017196999
 210104528283517112975405076841785849138031175816574935471201160734
 28308784094458566309376293495841288681579011970917457121134621266766
 41992780136134115607072325229498131612740008222722799275134941756283

Test Error Rate (%)

		Test Error Rate (%)
Linear classifier (1-layer NN)		12.0
K-nearest-neighbors, Euclidean		5.0
K-nearest-neighbors, Euclidean, deskewed		2.4
MNIST Digit Recognition		
- Handwritten digits 28x28 pixel images: $d = 784$ 60,000 training samples 10,000 test samples	K-NN, Tangent Distance, 16x16	1.1
	K-NN, shape context matching	0.67
	1000 RBF + linear classifier	3.6
	SVM deg 4 polynomial	1.1
	2-layer NN, 300 hidden units	4.7
Yann LeCunn	2-layer NN, 300 HU, [deskewing]	1.6
	LeNet-5, [distortions]	0.8
	Boosted LeNet-4, [distortions]	0.7



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What is the best distance metric between data points?

- Typically Euclidean distance
- Locality sensitive distance metrics
- Important to normalize.
Dimensions have different scales

How many K?

- Typically $k=1$ is good
- Cross-validation (try different k !)



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Distance metrics

$$D(\mathbf{x}, \mathbf{y}) = \sqrt{(x_1 - y_1)^2 + \cdots + (x_N - y_N)^2} \quad \text{Euclidean}$$

$$D(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|} = \frac{x_1 y_1 + \cdots + x_N y_N}{\sqrt{\sum_n x_n^2} \sqrt{\sum_n y_n^2}} \quad \text{Cosine}$$

$$D(\mathbf{x}, \mathbf{y}) = \frac{1}{2} \sum_n \frac{(x_n - y_n)^2}{(x_n + y_n)} \quad \text{Chi-squared}$$



Distance metrics

L1 (Manhattan) distance

$$d_1(I_1, I_2) = \sum_p |I_1^p - I_2^p|$$

L2 (Euclidean) distance

$$d_2(I_1, I_2) = \sqrt{\sum_p (I_1^p - I_2^p)^2}$$

- Two most commonly used special cases of p-norm

$$\|\mathbf{x}\|_p = (|x_1|^p + \cdots + |x_n|^p)^{\frac{1}{p}} \quad p \geq 1, \mathbf{x} \in \mathbb{R}^n$$



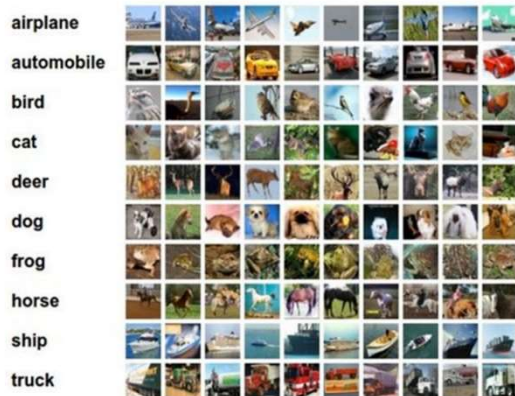
CIFAR-10 and NN results

Example dataset: **CIFAR-10**

10 labels

50,000 training images

10,000 test images.



For every test image (first column),
examples of nearest neighbors in rows

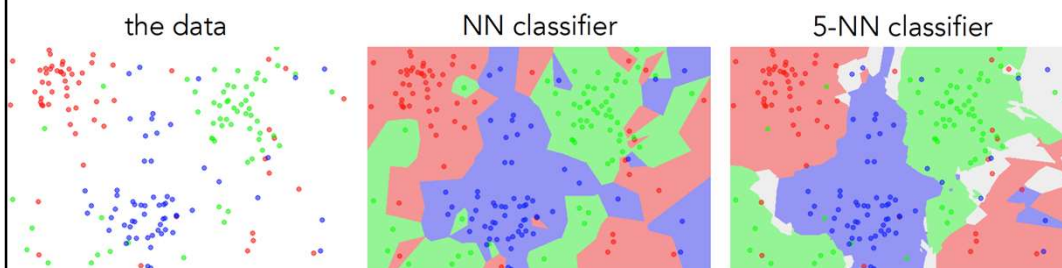


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k-nearest neighbor

- Find the k closest points from training data
- Labels of the **k points “vote”** to classify



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Hyperparameters

- What is the best distance to use?
- What is the best value of k to use?
- i.e., how do we set the hyperparameters?
- Very problem-dependent
- Must try them all and see what works best

Try out what hyperparameters work best on test set.

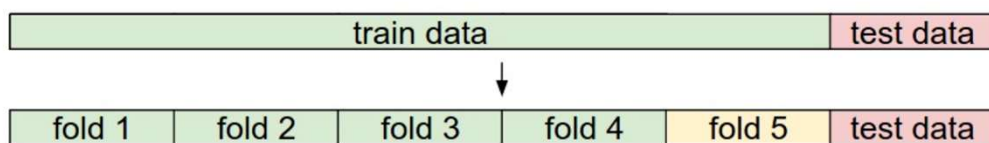


Trying out what hyperparameters work best on test set:

Very bad idea. The test set is a proxy for the generalization performance!
Use only **VERY SPARINGLY**, at the end.



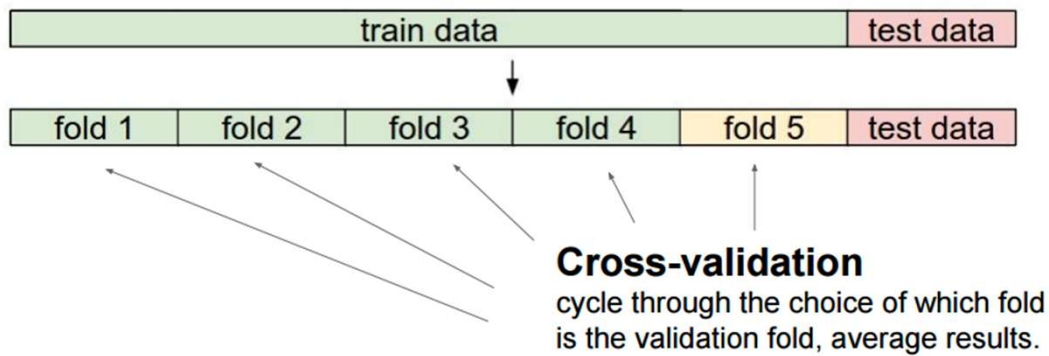
Validation



Validation data

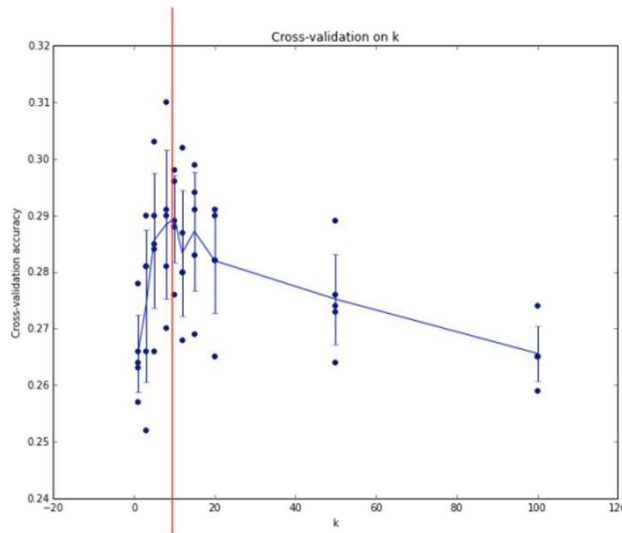
use to tune hyperparameters
evaluate on test set ONCE at the end

Cross-validation



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Example of 5-fold cross-validation for the value of k .

Each point: single outcome.

The line goes through the mean, bars indicated standard deviation

(Seems that $k \approx 7$ works best for this data)



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How to pick hyperparameters?

- Methodology
 - Train and test
 - Train, validate, test
- Train for original model
- Validate to find hyperparameters
- Test to understand generalizability



kNN

Pros

- simple yet effective

Cons

- search is expensive (can be speed-up)
- storage requirements
- difficulties with high-dimensional data



kNN -- Complexity and Storage

- N training images, M test images
- Training: $O(1)$
- Testing: $O(MN)$
- Hmm...
 - Normally need the opposite
 - Slow training (ok), fast testing (necessary)



Other classifiers

- Naïve Bayes
- SVM
- Random Forest
- Neural Network
- ...



References

Most of these slides were adapted from:

1. Ioannis Yannis, Gkioulekas (16-385 Computer Vision, Spring 2020, CMU)
2. Kristen Grauman (CS 376: Computer Vision, Spring 2018, The University of Texas at Austin)
3. Noah Snavely (Cornell University)
4. Fei-Fei Li (Stanford University)

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**VIỆN CÔNG NGHỆ THÔNG TIN VÀ TRUYỀN THÔNG**
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Thank you!

soict.hust.edu.vn/ fb.com/groups/soict